

Jegadeesh D

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PROFILE

Engineer with hands-on experience building end-to-end ML, deep learning, and RAG systems. Delivered 3 production-grade projects — NLP, computer vision, GenAI — with FastAPI, Docker, GCP Cloud Run, and measurable metrics. Seeking a role to drive real-world impact through data pipelines, model development, and scalable deployment.

TECHNICAL SKILLS

Languages and Database — Python, SQL, Pandas, NumPy, PostgreSQL, SQLite, SQLAlchemy, OpenCV

Machine Learning and Deep Learning — scikit-learn, XGBoost, Gradient Boosting, Random Forest, TF-IDF, Feature Engineering, Keras, TensorFlow, PyTorch, EfficientNetB0, Transfer Learning, CNN, Grad-CAM

NLP & GenAI — RAG, LLM, Sentence Transformers (MiniLM), ChromaDB, Vector Database, OpenRouter API

Deployment & Tools — FastAPI, Streamlit, Docker, GCP Cloud Run, Hugging Face Hub, GitHub Actions, pytest, Git

PROJECTS

Financial Intelligence Copilot

- Architected a dual-domain RAG pipeline over RBI/SEBI regulatory PDFs and HDFC annual reports, chunking 3,459 semantic segments via PyMuPDF with paragraph-aware splitting (800-char / 100-char overlap).
- Engineered vector search using Sentence Transformers (MiniLM) + ChromaDB, achieving 90% top-5 retrieval hit rate (9/10 eval queries) against a $\geq 70\%$ target benchmark.
- Exposed a production FastAPI endpoint (POST /ask) with confidence gating and page-level citations; packaged with Docker and validated across 76 pytest test cases.

AI-Powered Rice Leaf Disease Detection System

- Built a 4-class CNN image classifier using EfficientNetB0 transfer learning (Keras 3 + PyTorch) on the Mendeley rice leaf dataset, achieving 98.66% test accuracy across Bacterial Blight, Rice Blast, Brown Spot, and Tungro classes.
- Integrated Grad-CAM (Gradient-weighted Class Activation Mapping) explainability to highlight disease-triggering leaf regions, improving model transparency for agricultural domain experts.
- Deployed FastAPI inference API (POST /predict) to GCP Cloud Run, published model to Hugging Face Hub, and automated CI/CD releases via GitHub Actions; validated with a pytest API test suite.

Customer Support Tickets Analytics

- Designed a 3-task ML data pipeline over ~200K support tickets combining TF-IDF text features with structured metadata via scikit-learn ColumnTransformer; achieved 82.1% priority classification accuracy and $R^2 = 0.72$ for resolution time regression.
- Applied class-imbalance handling (class_weight='balanced'), multi-model comparison (Gradient Boosting, Random Forest, MLP), and holdout validation for leakage-free, production-defensible metrics.
- Deployed FastAPI inference endpoints (POST /predict_priority) with PostgreSQL backend, a multi-page Streamlit ops dashboard, and pytest API regression tests.

CERTIFICATIONS

Master Data Science Program by HCL GUVI 📄 — Feb 2026-Jun 2026

EDUCATION

MTech (integrated) in Software Engineering

Vellore Institute Of Technology
CGPA -7.89 / 10

July 2020 –
September 2025
Chennai

LEADERSHIP

Photography club, VIT Chennai

Advisory board

Mentored 70+ students on event execution and team coordination

January 2024